

Fall 2025 CPTS530 Homework 4  
MATH 448-548  
Numerical Analysis

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# Problem 1

## Problem Statement (Textbook 4.1.7):

Let  $A$  have the block form

$$A = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

in which the blocks are  $n \times n$ . Prove that if  $(B - I)$  is nonsingular, then, for  $k \geq 1$ ,

$$A^k = \begin{bmatrix} B^k & (B^k - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

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**Solution:**

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## Problem 2

**Problem Statement (Textbook 4.1.12):**

Let  $A$  be an  $n \times n$  invertible matrix, and let  $u$  and  $v$  be two vectors in  $\mathbb{R}^n$ . Find the necessary and sufficient conditions on  $u$  and  $v$  in order that the matrix

$$\begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$$

be invertible, and give a formula for the inverse when it exists.

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**Solution:**

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## Problem 3

### Problem Statement (Textbook 4.1.17):

A square matrix  $A$  is said to be **skew-symmetric** if  $A^T = -A$ . Prove that if  $A$  is skew-symmetric, then  $x^T Ax = 0$  for all  $x$ .

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**Solution:**

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## Problem 4

### Problem Statement (Textbook 4.1.18):

(Continuation) Prove that the diagonal elements of a skew-symmetric matrix are **0**. Also, prove that the determinant is 0 when the matrix is of odd order.

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**Solution:**

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## Problem 5

### Problem Statement (Textbook 4.1.19):

(Continuation) Let  $A$  be any square matrix, and define  $A_0 = \frac{1}{2}(A + A^T)$  and  $A_1 = \frac{1}{2}(A - A^T)$ . Prove that  $A_0$  is symmetric, that  $A_1$  is skew-symmetric, that  $A = A_0 + A_1$  and that for all  $x$ ,  $x^T A x = x^T A_0 x$ . This explains why, in discussing quadratic forms, we can confine our attention to **symmetric matrices**.

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**Solution:**

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## Problem 6

**Note:** This section contains Problem 6 from the homework, which corresponds to Textbook 4.2.1 with three parts (a, b, c). Each part is presented separately below.

**Problem 6(a)****Problem Statement (Textbook 4.2.1a):**

Prove these facts, needed in the proof of Theorem 2.

(a) If  $U$  is upper triangular and invertible, then  $U^{-1}$  is upper triangular.

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**Solution:**

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**Problem 6(b)****Problem Statement (Textbook 4.2.1b):**

Prove these facts, needed in the proof of Theorem 2.

(b) The inverse of a unit lower triangular matrix is unit lower triangular.

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**Solution:**

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**Problem 6(c)****Problem Statement (Textbook 4.2.1c):**

Prove these facts, needed in the proof of Theorem 2.

(c) The product of two upper (lower) triangular matrices is upper (lower) triangular.

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**Solution:**

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## Problem 7

### Problem Statement (Textbook 4.2.5):

Prove that an upper or lower triangular matrix is nonsingular if and only if its diagonal elements are all different from 0.

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**Solution:**

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# Appendix